

FULL-STACK PORTFOLIO PROJECT

# Real-Time Chat Application

using Spring Boot, WebSocket, React & PostgreSQL

React.js

Spring Boot

WebSocket

PostgreSQL

JWT

Email OTP

A secure real-time chat web application with user registration, email verification, login, online user presence, direct messaging, media messages, profile settings, theme selection, group chat, admin controls and polls.

Corrected tech stack version - no MERN / Node.js / Express.js / MongoDB / Socket.IO wording

## What is this project?

This is a full-stack real-time chat application built using React.js frontend and Spring Boot backend. WebSocket is used for instant message delivery and online presence updates. PostgreSQL stores user accounts, chats, groups, messages, polls and profile data. JWT authentication secures user sessions.

## Registration and login flow

Users sign up by entering username, display name, email and password. After submitting, an OTP is sent to the entered email and the user is redirected to the Verify OTP page. After successful verification, the account becomes active and the user can log in securely using JWT.

## Real-time messaging

- Direct user-to-user chat
- Instant WebSocket updates
- Message history
- Edit and delete messages

## Rich chat features

- Text messages
- Emojis and mentions
- Photo and video sharing
- Online/offline status

## Profile and theme

- Update display name
- Upload profile picture
- Select chat themes
- Secure sign out

## Group communication

- Create groups
- Assign admins
- Remove members
- Group settings and members

## Polls and collaboration

- Create polls inside groups
- Group member participation
- Navigate to user inboxes
- Organized chat workspace

## Security and backend

- JWT authentication
- Email OTP verification
- Spring Boot REST APIs
- PostgreSQL database

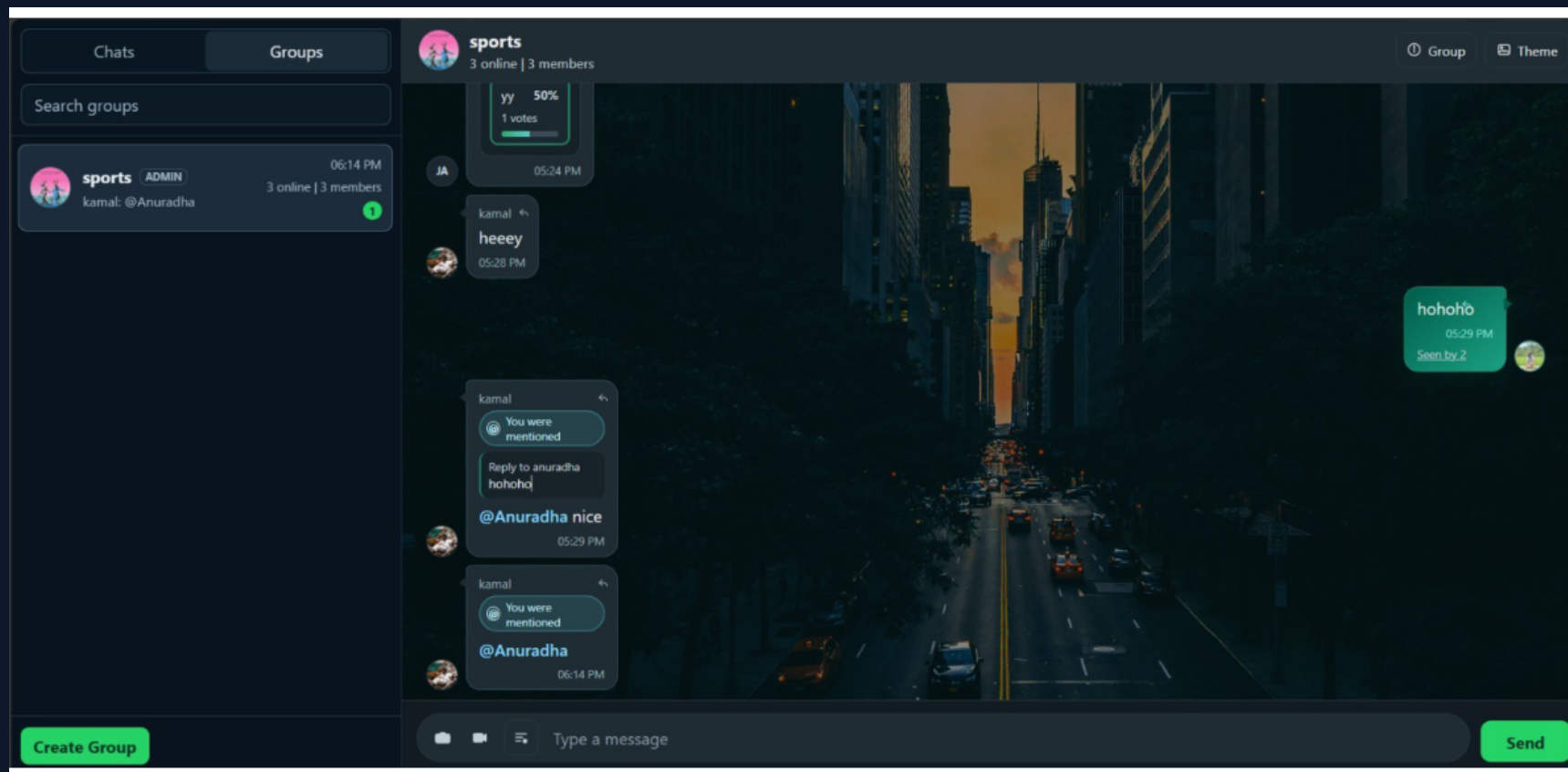
The screenshot displays three distinct authentication screens for a chat application, each with a dark theme and light-colored text and input fields.

- Login Screen:** Features the title "Login" and the instruction "Continue to your chat workspace." It includes input fields for "Username" and "Password", a "Show Password" toggle, a prominent green "Login" button, a "Forgot Password?" link, and a "No account? Create one" link.
- Register Screen:** Features the title "Register" and the instruction "Create your account and start chatting." It includes input fields for "Display Name", "Username", "Email", "Password", and "Confirm Password" (with a "Repeat password" sub-field). It also has "Show Password" and "Show Confirm Password" toggles, a prominent green "Create Account" button, and a link for "Already have an account? Login".
- Verify Email Screen:** Features the title "Verify Email" and the instruction "Enter the OTP sent to your email." It includes input fields for "Email" (pre-filled with "anuradhawork123@gmail.com") and "OTP", a "Show Password" toggle, a prominent green "Verify" button, a "Resend OTP" button, and a "Back to Login" link.

## Authentication Flow - Login, Register and Verify Email

This screenshot explains the main account flow. Users can log in, create a new account, and verify their email using OTP before accessing the chat workspace.

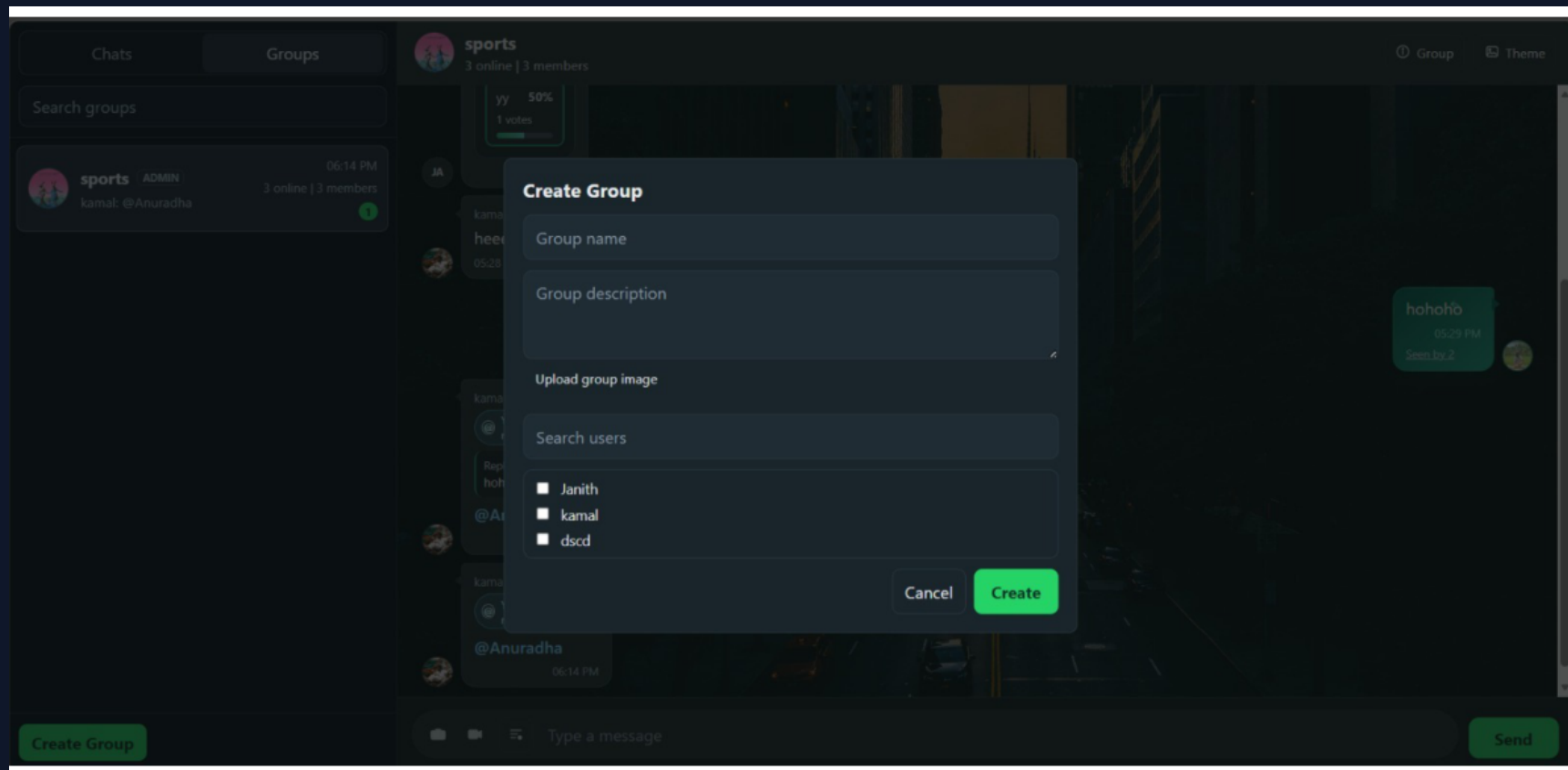
**Key items:** Login page, Register page, Email OTP verification



## Group Chat Workspace

This screen shows the group chat workspace where users can communicate inside groups while still using the chat sidebar and real-time message area.

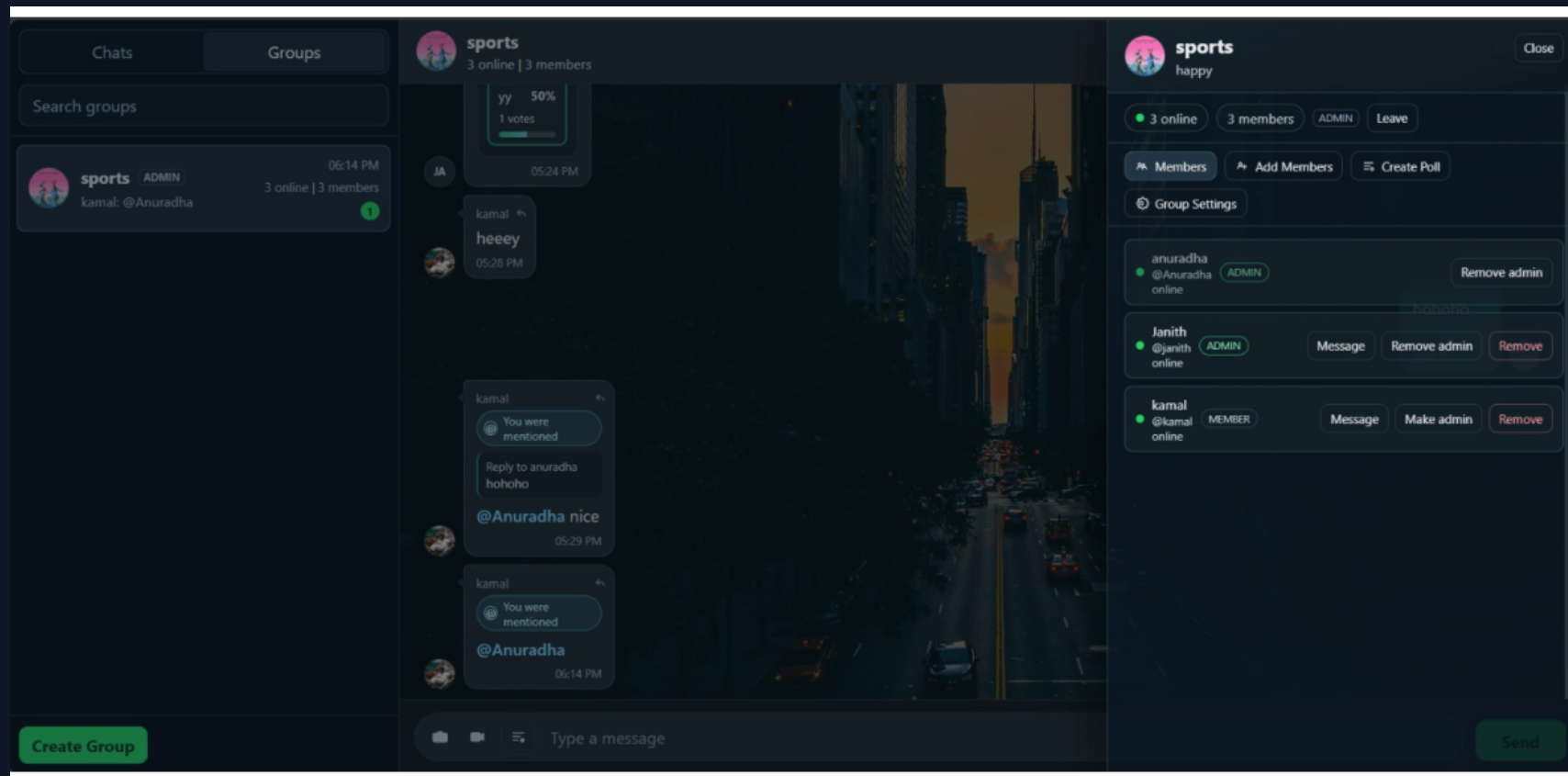
**Key items:** Group chat, Real-time updates, Sidebar navigation



## Create Group Dialog

Users can create a new group, add a group name, choose members, and start a group conversation.

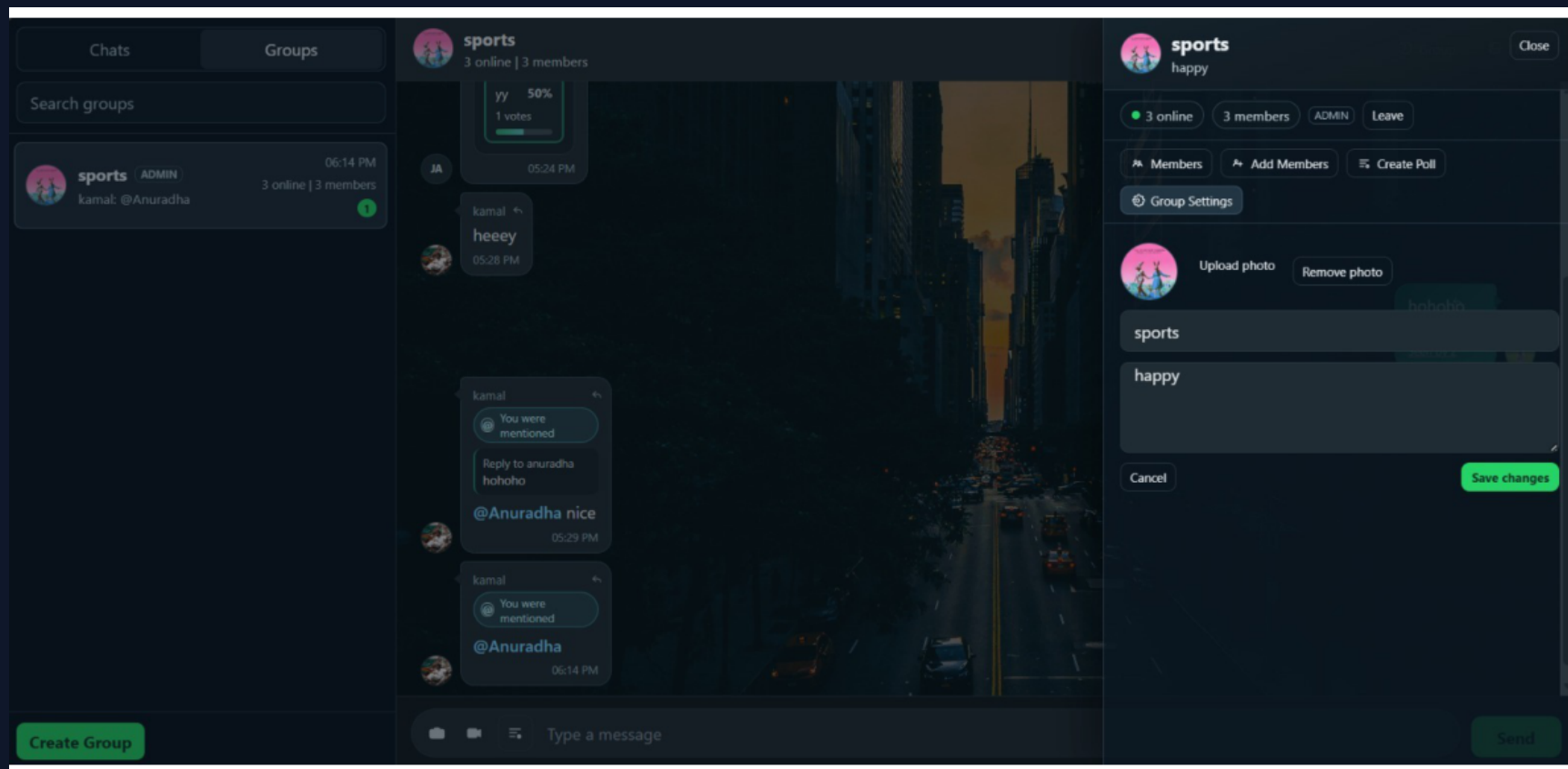
**Key items:** Create group, Member selection, Group setup



## Group Members and Admin View

This view shows group members and admin controls. The creator or admin can manage group membership and roles.

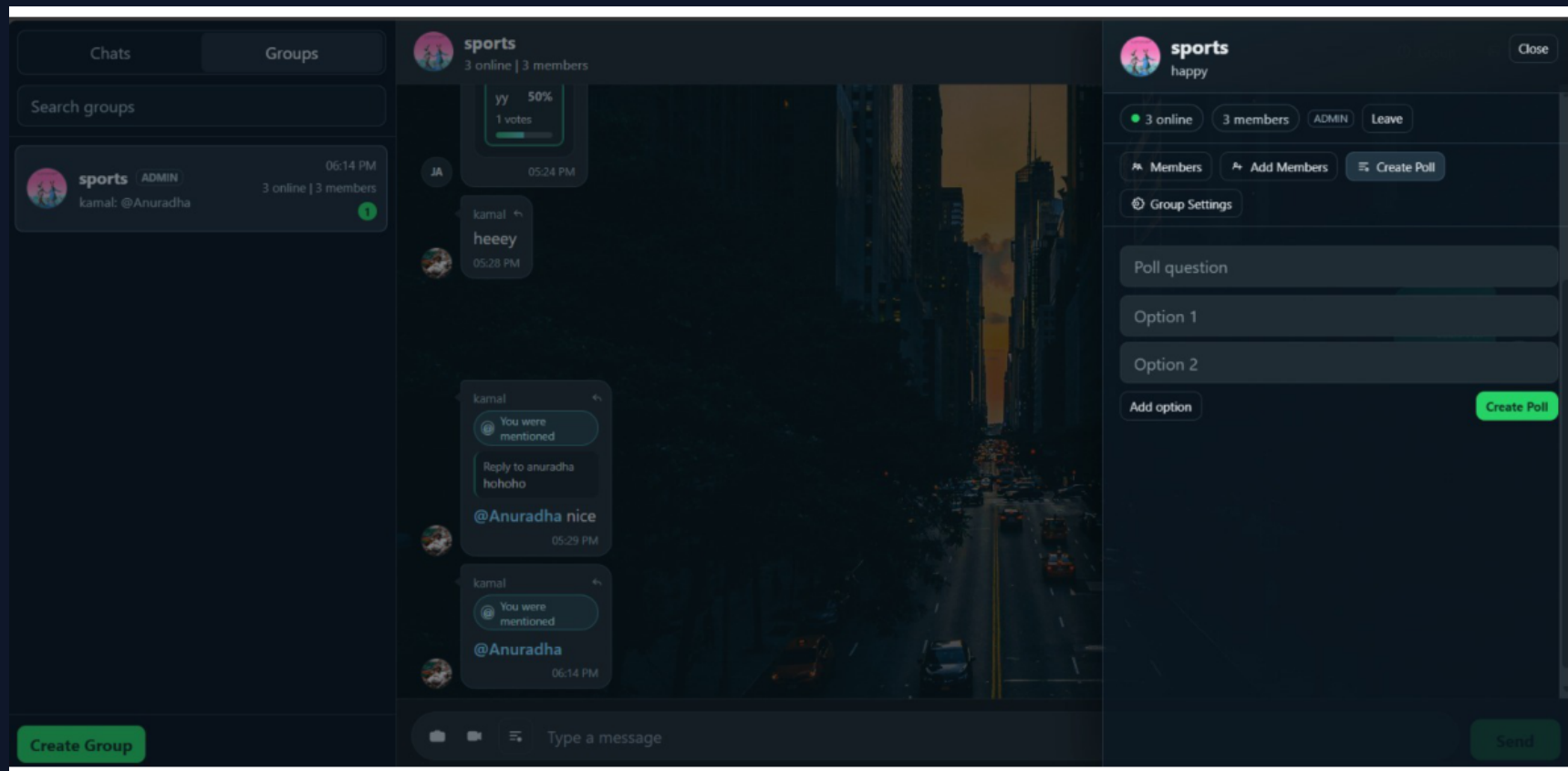
**Key items:** View members, Admin role, Member management



## Group Settings and Management

Group settings allow admins to update group details, manage members, and control group actions.

**Key items:** Group settings, Remove members, Admin controls

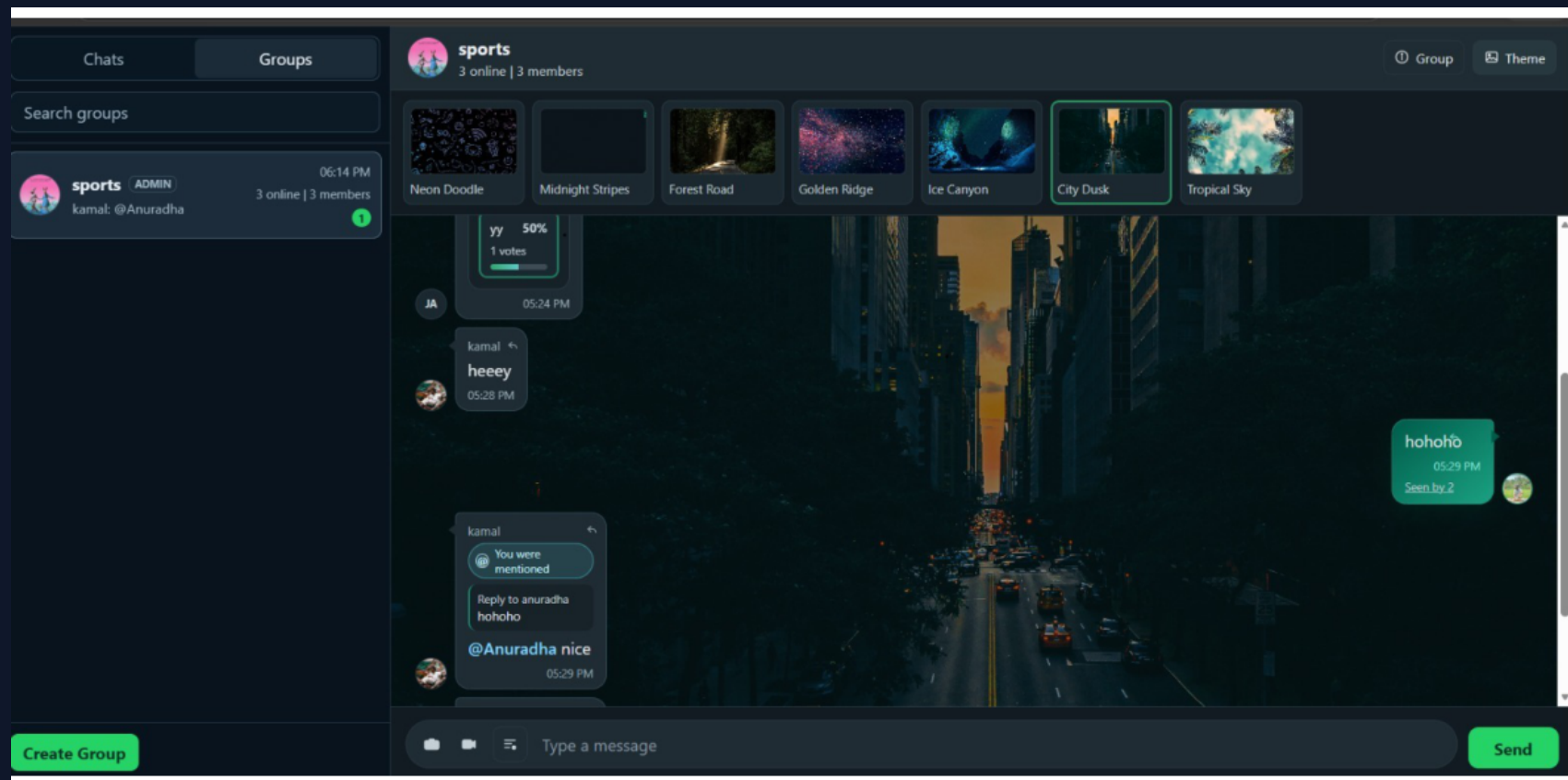


## Poll Creation Feature

Users can create polls inside group chats to collect opinions from group members.

**Key items:** Create poll, Group collaboration, Poll options





## Media Sharing in Chat

The chat interface supports rich message sharing including images, photos or other media inside conversations.

**Key items:** Media message, Image sharing, Chat history